The Climate Seesaw 🌍

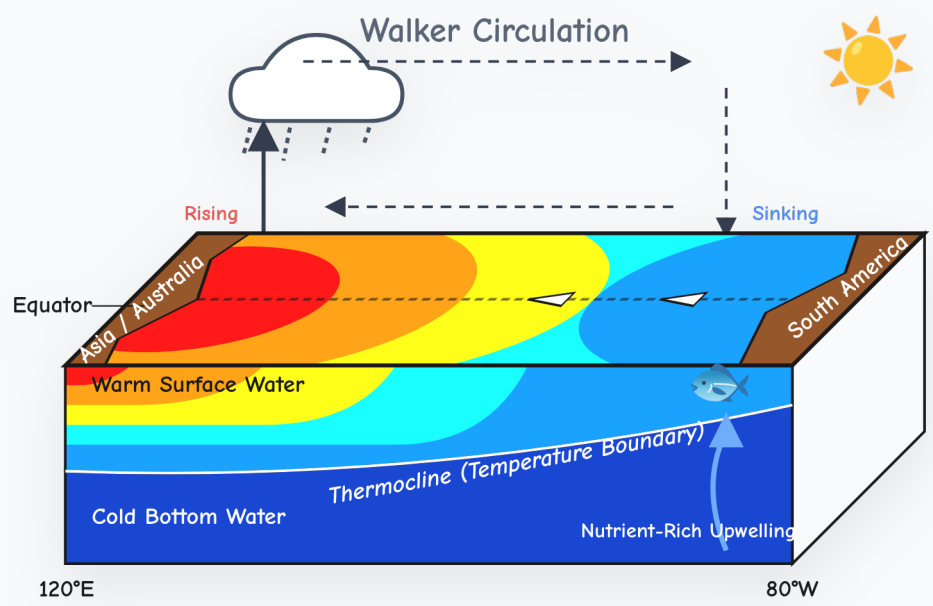
Understanding El Niño & La Niña

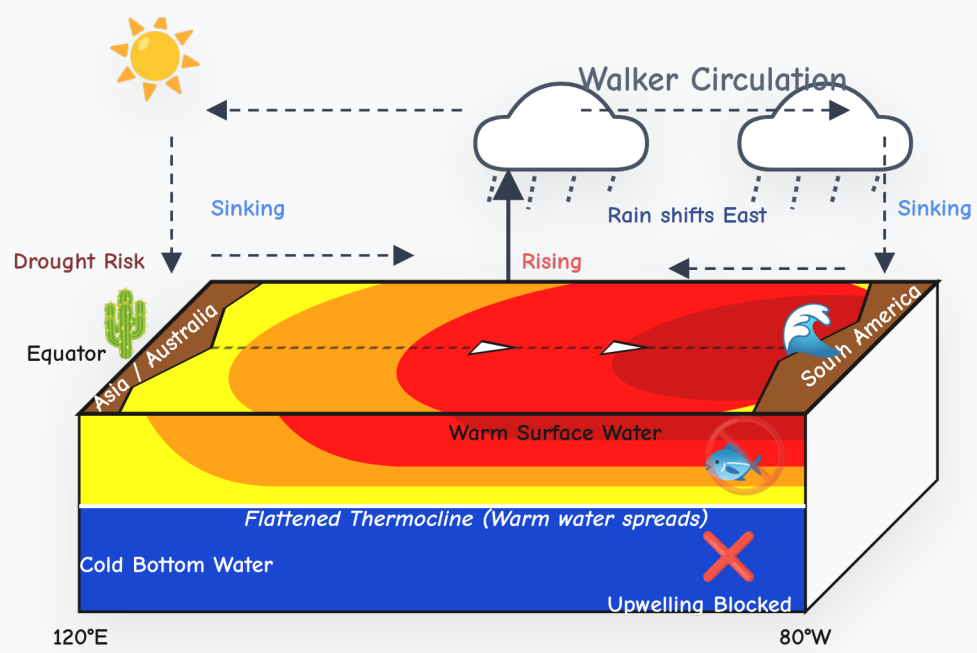
The **El Niño-Southern Oscillation (ENSO)** is a naturally occurring cycle of warm and cold ocean temperatures in the Pacific Ocean. It swings back and forth like a seesaw every 2-7 years, drastically changing wind patterns, ocean circulation, and weather across the entire globe!

1. Neutral Conditions 😌

Under normal conditions, the Pacific Ocean behaves predictably:

- Steady **Trade Winds** blow from East to West.
- Warm water piles up in the West (near Asia).
- Rising warm air brings heavy 🌧️ **rain** to the West.
- In the East (South America), deep, cold water rises to the surface. This **upwelling** brings nutrients, supporting massive 🐟 fish populations!



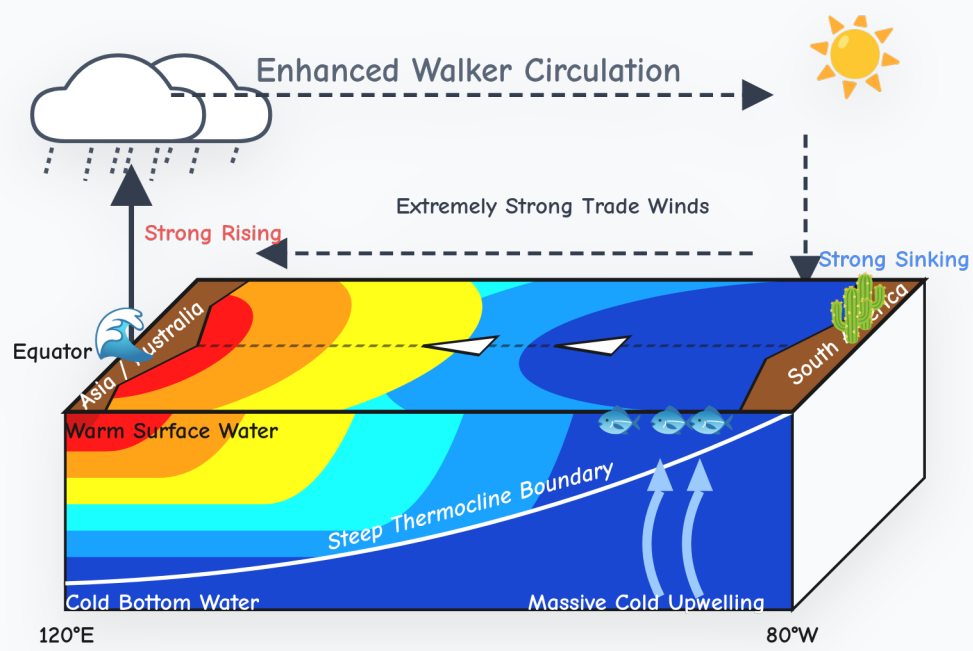


2. El Niño (Warm Phase) 🔥

Everything flips during an El Niño event:

- Trade winds severely **weaken** or even reverse!
- The warm water pool "sloshes" back East.
- Rain clouds follow the warm water, causing **massive floods** 🌊 in South America.
- Australia and Asia suffer from severe **droughts** 🏜️.
- Upwelling stops completely, devastating the fishing industry.

3. La Niña (Cold Phase) ❄️

La Niña is essentially "Normal conditions on overdrive!"

- Trade winds blow **much stronger** than usual.
- Even more warm water is forcefully pushed to the far West. Massive amounts of cold water upwell in the East.
- This causes extreme **flooding and cyclones** ⛈️ in Australia & Asia (**Strong Monsoon**).
- The Americas experience unusually dry weather 🌵 and strong droughts.